SIMATS ENGINEERING

ASSESSMENT TOOL 2: Technical Presentation

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

Name	**M.SANDEEP**
Register Number	**192525095**

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 50**
Question Count: 5

Code of Conduct

I M.Sandeep(192525095) (Name / Reg. No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: ___M.Sandeep_________________

Assessment Tasks

Technical Presentation Topics

Prepare a 6-8 minute presentation on any one of the following topics. Your slides must include definitions, architecture, industry relevance, and one practical example.

1. Cloud computing characteristics and benefits for startup ecosystems.
2. How virtualization enabled the growth of cloud service providers.
3. A comparative presentation on IaaS, PaaS, and SaaS for an educational institution.
4. Web services in cloud platforms: SOAP vs REST for enterprise integration.
5. Why XaaS is becoming a dominant consumption model in modern IT.

Minimum slide flow:

- Title and objective
- Core technical explanation
- Architecture or process diagram
- Real-world use case
- Conclusion and key takeaway

Technical Presentation Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Technical Content	30%	Content is highly accurate, relevant, and insightful	Content is correct with minor gaps	Basic content with limited depth	Content is inaccurate or superficial
Structure and Flow	20%	Excellent logical flow and slide organization	Good structure with minor issues	Some organization but weak flow	Disorganized presentation
Use of Visuals	15%	Visuals strongly support technical communication	Relevant visuals used well	Limited or unclear visuals	No meaningful visuals
Communication Skills	20%	Confident, clear, and engaging delivery	Clear delivery with minor hesitation	Moderate clarity but uneven delivery	Weak delivery and low clarity
Professionalism	15%	Well-prepared and audience-focused presentation	Mostly professional	Basic preparation visible	Poor preparation and professionalism

TITLE:
COMPARATIVE ANALYSIS OF IAAS, PAAS, AND SAAS FOR AN EDUCATIONAL INSTITUTION

CSA1522-Cloud computing & Big Data Analytics

M. Sandeep(192525095)
M. Jaya Kirshna(192524500)
G. Venkata Niranjan Reddy(192525199)

Guided by
RAJARAM PICHAMUTHU

INTRODUCTION TO CLOUD COMPUTING

- Cloud computing provides computing resources over the internet.
- It reduces the need for institutions to maintain physical IT infrastructure.
- Educational institutions can use cloud services for teaching, learning, and administration.
- The three major service models are **IaaS, PaaS, and SaaS**.

IAAS – INFRASTRUCTURE AS A SERVICE

- IaaS provides virtualized **servers, storage, and networking**.
- The institution manages the operating system and applications.
- It offers high flexibility and control over infrastructure. Resources can be increased or decreased based on demand. Example: **Amazon EC2, Microsoft Azure Virtual Machines, Google Compute Engine**.

PAAS – PLATFORM AS A SERVICE

- PaaS provides a ready-made platform for application development.
- Developers do not need to manage physical servers.
- It includes development tools, databases, and runtime environments.
- It allows faster development and deployment of applications.
- Examples: **Google App Engine, Microsoft Azure App Service, Heroku**.

SAAS – SOFTWARE AS A SERVICE

- SaaS provides ready-to-use software through the internet.
- Users do not need to install or maintain the software. The cloud provider manages the application and infrastructure.
- It can be accessed from computers, tablets, and smartphones.
- Examples: **Google Workspace, Microsoft 365, Zoom**

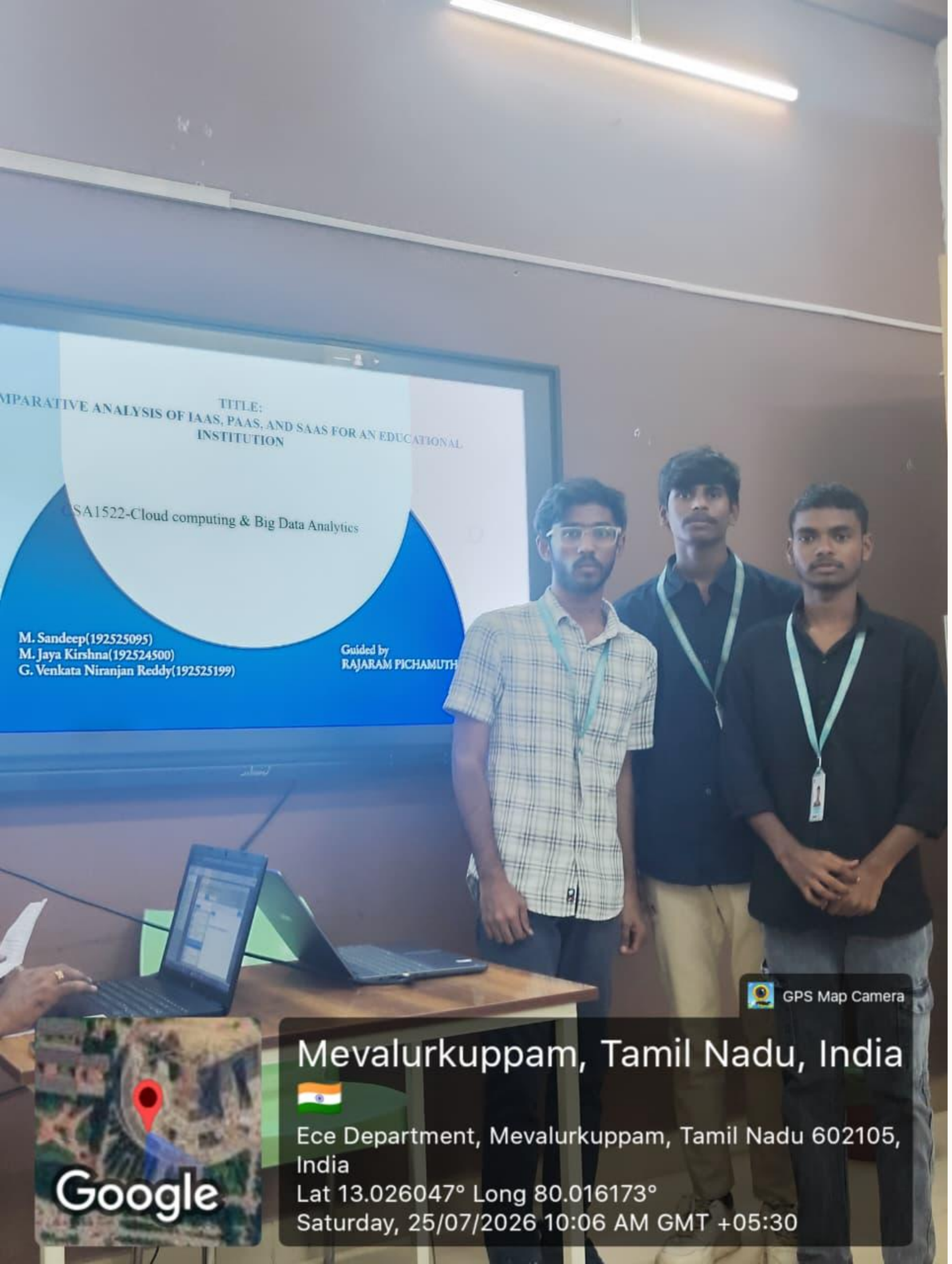
COMPARISON OF IAAS, PAAS, AND SAAS

Feature	IaaS	PaaS	SaaS
Provides	Infrastructure	Development Platform	Ready-to-use Software
User Control	High	Medium	Low
Management	IT Team	Developer	Service Provider
Educational Use	Hosting systems	Developing applications	Online learning & collaboration
Example	AWS EC2	Google App Engine	Google Workspace

CONCLUSION

- **IaaS** is suitable for institutions needing high infrastructure control.
- **PaaS** is suitable for developing and deploying custom educational applications.
- **SaaS** is ideal for everyday services such as email, online classes, and collaboration.
- A combination of all three models can meet different institutional requirements.
- Cloud computing improves **scalability, flexibility, accessibility, and cost efficiency**

THANK YOU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GPS Map Camera
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